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| Virtual University | **Assignment No. 02 (Graded)**<br>**SEMESTER Spring 2021**<br>**BIO503 – Biological Biophysics** | **Total Marks: 20**<br>**Start Date: 6.8.2021** |
| | | **Marks: 15** |

# Assignment topic

## New approaches to study macromolecular structure and function

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Macromolecules are made up of small basic molecular units. It includes the proteins which is polymers of amino acids, nucleic acids which is polymers of nucleotides, carbohydrates which is polymers of sugars and lipids with a variety of modular constituents. The biosynthesis and degradation of biological macromolecules involves linear polymerization, breakdown steps proteins, nucleic acids and lipids and may also involve branching or debranching of carbohydrates. These processes may involve multi-protein complexes e.g., ribosome, proteasome with complex regulation. Biophysics provides knowledge about the many macromolecules which deals them at a nanoscale. Covalent and non-covalent bonding govern the three-dimensional structures of proteins and nucleic acids which impacts function. The amino acid sequences observed in nature are highly selected for biological function but do not necessarily adopt a unique folded structure. Biophysics is the broadest definitions these terms refer to the science and technology that derives from being able to assemble, manipulate, observe and control matter on length scales from one nanometre up to 100 nanometres for instance the size of a protein molecule is 5 nm. Macromolecules interact with other molecules using a variety of non-covalent interactions. The specificity and affinity of these interactions are critical

to biological function. Some macromolecules catalyze chemical reactions or facilitate physical processes. Nanoscience provides important opportunities for theory and computation to lead in the discovery process because the experimental tools often provide an incomplete picture of the structure and function of nanomaterial. This theory can often fill in missing features crucial to understanding what is being measured. Biophysics approach to nanoscale can play such a useful role for micrometer-scale systems, are sometimes incapable of describing nanoscale properties due to incomplete incorporation of the underlying Biophysics in the size-dependent materials parameters. Biophysics that are having an undeniable impact in real life. For example, next generation sequencing which resides in our understanding of the Biophysics of the pairing and replication of DNA strands.

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